NCI
Coláiste Náisiúnta na hÉireann | National College of Ireland

This certificate is awarded to:

Mohd Nizam Mohd Nasir Shaikh

FOR COMPLETING THE PROGRAMME

Academic Integrity: For Students, Second Edition

January 15, 2026

Certificate Verification Code: SfYu3HrPhR

epigeum

Adaptive Portfolio Management in Cryptocurrency Markets: An Analysis of AI-Driven Deep Reinforcement Learning for Dynamic Assets and Transaction Costs

https://doi.org/10.1016/j.eswa.2020.114002

Mohd Nizam Mohd Nasir Shaikh
24198170 – X24198170@student.ncirl.ie
MSc in Fintech (MSCFTD1)
National College of Ireland

Keywords : Deep reinforcement learning; AI-driven trading; Recurrent neural networks (GRU); Portfolio management; Transaction costs

1. Background Information

1.1. Research Question

The paper studies how to build an automated trading strategy in the cryptocurrency markets where the number of tradable assets fluctuates over the time (Betancourt & Chen, 2021). Traditional deep reinforcement learning (RL) portfolio models typical assume a set list of assets which makes it difficult to include newly-listed coins without disruptive redesign and retaining of the model. In this paper, AI-driven trading refers to the application of deep reinforcement learning agent instead of predefined rules by humans in determining how to distribute the cryptocurrency portfolio using directly available market data. The main research question is: Can a deep RL based trading system manage a portfolio in such a dynamic market and explicitly manage transaction costs plus able to continuously outperform existing strategies in terms of return and risk?

1.2. Specific Task

The papers specific task can be performed as online portfolio management on a large cryptocurrency exchange that has hundreds of cryptocurrencies and often lists new coins. At each decision time, RL agent takes a look at current market information of all currently available assets and determines how to rebalance wealth between cash and various cryptocurrencies. Firstly, system creates target portfolio vector which indicates the desired weight of each asset including cash. Secondly, it calculates which assets to buy or sell to get from current portfolio to target and adhering to asset specific transaction fees. The authors evaluate the approach in 3 back testing scenarios involving varying lengths of episode (1 day, 30 days and 16 weeks) and different holding periods (30 minutes, 1 day and 1 week). Thus, able to view the behaviour of the strategy in the context of short-term, medium-term and long-term crypto trading (Betancourt & Chen, 2021).

1.3. Methods

The methodology is a model-free deep RL framework using Proximal Policy Optimization (PPO) in actor-critic architecture, where the asset states for each market which are modelled as a tensor of the size $n \times f \times k$ (assets $\times$ features $\times$ look-back steps) that include prices, volumes and current portfolio weights. A recurrent neural network with Gated Recurrent Units (GRUs) processes each time series for each asset and the outputs are combined using layers weighted by current portfolio shares. This method makes the architecture independent of their number of assets, since new coins can be added without the need to change the network structure. In this framework, the deep RL agent replaces hand crafted trading rules: GRU based policy network which has been trained using the Policy Optimization (PO) algorithm, which gets the help by learning form the simulated market interaction on how a desired portfolio allocation can be mapped from past prices, volumes and positions than passed to an optimization layer sensitive to transaction costs. This optimization layer represented as a linear program that determines how much each asset to purchase or sold to achieve the desired portfolio while maximizing its value after trades while heterogenous fees are modelled, cash flows and transaction costs in model represented by linear constraints. Overall, the agent is trained in various parallel simulated environments using PPO and generalized advantage estimation with such rewards derived from periods returns using a Sharpe style signal. In order to ensure that the policy learns a balance between profitability and risk over the course of a full trading episode.

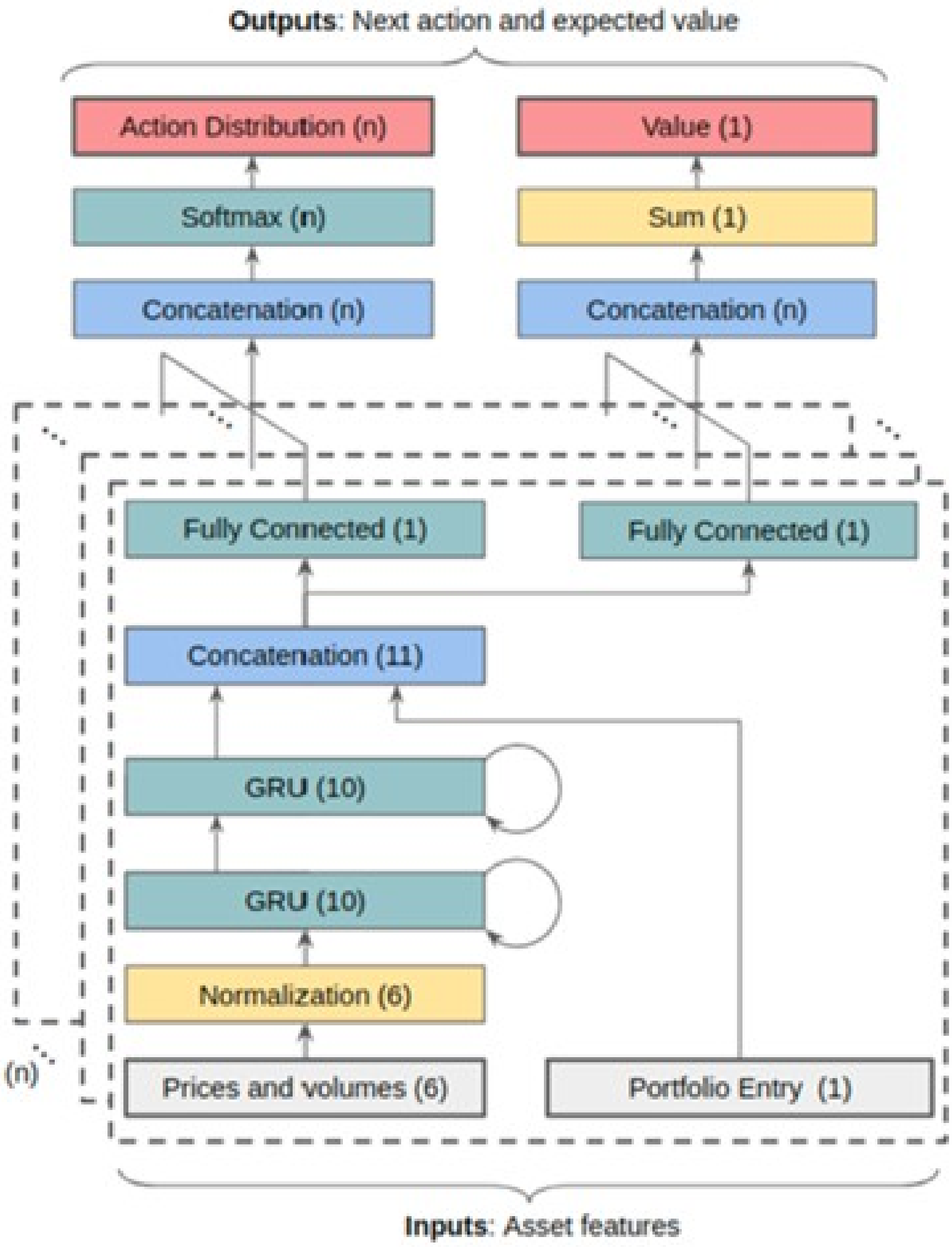

Figure 1: Outline of the proposed Deep Reinforcement Learning (DRL) architecture for portfolio management processing assets independently (Betancourt & Chen, 2021).

Table 1: Input features used in the state tensor for each asset (Betancourt & Chen, 2021).

Feature	**Description**
Close price	Normalized closing price at the end of period
High price	Normalized highest price during the period
Low price	Normalized lowest price during the period
Volume	Normalized volume traded during the period
Portfolio weight	The current weight of the asset in portfolio

1.4. Key Findings

The authors test their method using historical data from 1 of the biggest cryptocurrency markets worldwide. Three configurations are analyzed: one day episodes with half hour holding, thirty-day episodes with daily rebalancing and sixteen-week episodes with weekly rebalancing. In terms of total return and Sharpe ratio, the suggested dynamic asset RL strategy has outperformed a number of benchmark trading techniques designed for fixed sets of assets across all 3 scenarios (Betancourt & Chen, 2021). Because it applies to each asset independently and merges them through the weights of a portfolio rather than as fixed position in an input grid. The results demonstrates that the architecture can trade coins that haven't happened in the training data. When it's compared to basic optimizations that are unaware of varied charge structures, transaction cost aware optimization improves operational performance and minimizes unnecessary portfolio churning. When considered collectively, these results suggests that the experiments are substantially achieves its goal which is an effective portfolio management on a dynamic set of cryptocurrency assets with the reasonable costs.

2. Critical Analysis

2.1. Contributions

The paper's primary contribution is a methodology for managing portfolios that was created specifically for markets with fluctuating asset values, a crucial aspect of cryptocurrency exchanges. The architecture may consist of all the available assets and seamlessly integrate with new listings upon deployment of time without changing the architecture by processing the assets independently and merging them using the portfolio method weighing the assets. Compared to approaches where a defined universe is hardcoded and only a tiny subset of assets may be processed which describes as a significant improvement. The intimate integration of transaction cost optimization into the RL decision loop is secondary significant addition. As the objective shifts from just matching desired weights in portfolio to maximizing the value of portfolio after expenses when the rebalancing phase is modelled as a linear program that includes fees for purchasing and selling assets. Altogether, these contributions offer a trading system that is more flexible than the conventional RL portfolio models and more in line with the actual exchange conditions.

2.2. Novelty

The approach in novel is how it handles the execution layer as well as market representation. As authors utilize a per asset recurrent feature extractor plus a portfolio weighted aggregation layer instead of using convolutional networks based on a fixed "image" of assets and time. This allows us to use entire market for training and makes the model size independent of the number of assets. In marketplaces like cryptocurrency, where assets come and go this style is particularly appropriate for use. In this paper, rebalancing has been defined as linear program that specifically takes into account for different buy and sell costs for each asset and directly optimizes the value preserving effect of trades. In contrast, many previous studies either ignore transaction costs or use a uniform cost of transactions regardless of asset. AI-driven portfolio management has significantly advanced with the integration of a realistic, optimization-based transaction module and a dynamic universe RL architecture.

2.3. Strength

The study illustrates a number of significant strengths, first, it is well adapted to the circumstances of actual cryptocurrency market as the data comes from a large exchange that is operated 24/7 where there's a lot of new listings and high volatility. As a result, environment of experiments is similar to the region where the method could be applied to solve the issue. Second, the evaluation is rather comprehensive, making use of three-time horizons and standard performance metrics like the Sharpe ratio and total return, which aid to understand the strategy's success under various trading frequencies. Third, since the input dimension depends only on the number of features rather than the number of assets, architecture is sample efficient and scalable allowing model to learn from all accessible assets rather than just being limited to a few. Lastly, the authors precise handling of transaction costs through linear programming demonstrates their knowledge of real-world trading limitations and their search of techniques that might be logically applied rather than merely theoretical models.

2.4. Weakness

At the same time, paper has a number of weaknesses too that makes it limited in its applicability and generality. Since each experiment is carried out on a single cryptocurrency market, it is unclear how the strategy would work in other asset classes with distinct microstructure, regulations and liquidity patterns for example bonds, fiat currency, stocks and foreign exchange. Order book depth, slippage, latency and market effect are examples of real execution frictions that are not fully taken into account in back tests. These frictions can significantly lower realized returns during actual live strategy deployment. The paper lacks the interpretability and model risk analysis required in institutional contexts for risk management and regulatory approval plus model is a complex completion recurrent neural network. Lastly, despite of the architectures ability to incorporate new assets at runtime as the study does not systematically investigate how resilient trained agent is to severe regime changes which are also common in cryptography and can result in crashes or protracted bear marks that could cause the system to behave sensitively.

2.5. Drawbacks

The actual work points out at several more general limitations of AI techniques, particularly in deep RL for this type of portfolio management task. Deep RL models are prone to overfitting since they require a lot of data because many cryptocurrencies have brief histories and reflect noisy behaviour, agent may pick up patterns that don't hold true outside of sample. Although the methodology offers point portfolio recommendations without a measure of uncertainty, traders ad risk managers are unable to determine when the model is making sound choices and when it may be fragile. Continually adjusting to a market that is changing quickly may require frequent retraining or fine tuning which is operationally taxing and could cause instability if steps are not taken to carefully regulate the retraining process. Furthermore, the models lack of transparency makes it hard to fully understand and accept situations when having a clear justification for actions is crucial.

2.6. Challenges

Considering this study in a larger context of AI in the financial sector highlights further issues. There is a lot of conflict between speed and transparency, although RL based systems might be able to respond promptly to fresh information and hidden patterns in data, regulators, clients and internal control functions will find it difficult to understand them. Financial institutions must implement strict model-risk management frameworks and some models are more challenging to validate, monitor and stress test due to their complexity and unclear operation. Hence, strong data and infrastructure are also necessary for AI-driven trading systems since errors in software or data flows can have rapid financial consequences when models are put into production. Additionally, if similar AI tactics are widely utilized, some agent might react to market signals using the same strategies which could increase volatility during stressful times that could lead to herding and pro-cyclical behaviour. Subsequently, it is important to consider these governance and systematic problems in addition to any potential performance improvements.

2.7. Frontier

In dynamic, high dimensional markets this paper is at close to frontier of AI-driven portfolio management. It goes beyond many other studies that are restricted to fixed tiny asset universes and simplified trading assumptions by taking into consideration the dynamic world of assets and accounting for realistic transaction costs. However, it also raises significant unanswered problems about accuracy, transparent risk and capital restrictions, cross-asset portfolios and regulatory compliance. Therefore, the work can be regarded as an important step towards more adaptive and realistic AI-driven trading systems but not yet a complete solution for institutional deployment. In future, this will probably expand upon by combining dynamic universe RL with formal risk controls, explainable decisions layers and multi-market coverage pushing it further towards institutional grade AI portfolio managers.

3. Summary

3.1. AI's Transformative Impact

This research shows how deep reinforcement learning can significantly change the trade strategy in context of managing cryptocurrency portfolios. The authors let an agent learn directly from market data on how to assign weight to a number of coins in a universe that is changing over the period, as opposed to manually prescribing rules. In this environment, a transaction cost aware optimization layer works with a deep reinforcement learning agent although it is implemented as GRU-based neural network trained via PPO to participate in digitization decisions by making and carrying out trades in a real-time asset universe. The method may be able to take advantage of early possibilities with new coins that traditional models would overlook because it can automatically include newly listed assets. AI techniques can be made to respect more than just idealized trading conditions, as demonstrated by addition of transaction aware optimization layer. Overall, the research indicates that while AI-driven portfolio managers may offer traders and risk teams more adaptable and responsive solutions for the cryptocurrency industry and it also highlights the importance of having robust controls, monitoring and validation.

3.2. Future Performance Directions

There are various possibilities that can lead to several directions we can take to improve performance, probabilities and practical application in the field of this work. Instead of focusing just on return and Sharpe ratio, one major approach is to create risk aware RL objectives that explicitly incorporate drawdown restrictions, tail risk metrics or regulatory capital measures. Another one is to incorporate uncertainty aware decision making such with the use of ensembles or Bayesian extensions, which provide portfolio recommendations together with data on robustness or confidence in various circumstances. An extension in taking the architecture to multi-asset portfolios encompassing not only crypto but also stocks, fiat currency, foreign exchange and other assets would test the approach universality and make it more relevant for diverse investors. Lastly, understanding explainability and stress testing from the development cycle such as analysing which features contribute to decisions, how does the policy respond in crisis scenarios that are simulated? Would aid in bridging the gap between highly effective black box models with transparency and governance that required of regulated financial organizations.

4. References

Betancourt, C. and Chen, W.-H. (2021) 'Deep reinforcement learning for portfolio management of markets with a dynamic number of assets', *Expert Systems with Applications*, 164, p. 114002. https://doi.org/10.1016/j.eswa.2020.114002.